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RESEARCH ARTICLE

Temporal State-Aware Task Replanning for Multi-UAV Coordination

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ABSTRACT Multi-UAV cooperative systems face adaptability challenges under dynamic task updates and node failures. Conventional replanning methods ignore the coupling between flight trajectories and task sequences, resulting in high computational overhead and limited stability. This work proposes a temporal state-aware multi-UAV cooperative task replanning framework. A hybrid centralized-distributed architecture with dynamic clustering is constructed to support stable swarm coordination, and a four-level hierarchical incremental replanning strategy is developed to implement adaptive local task adjustment. Experimental results show that the proposed method improves system task utility by 7.0%, reduces the number of participating UAVs by 16.7% during dynamic adjustment. The framework maintains stable performance under compound disturbances of new task addition and UAV failure, effectively balancing scheduling performance and computational consumption. Our code is available at https://github.com/Agentyzu/TSM_CTR

INDEX TERMS Multi-UAV coordination, task replanning, dynamic clustering, incremental replanning.

I. INTRODUCTION

UAVs have become indispensable platforms for complex mission execution, yet they are inherently constrained by limited onboard energy, sensing, and computational resources [1]. A single UAV is capable of performing simple tasks via autonomous navigation and precise control, whereas its operational performance is severely restricted in complex and dynamic scenarios [2], [3]. To overcome such individual limitations, multi-UAV cooperative systems have been widely adopted, which distribute sensing, computation, and control resources across a swarm to improve system reliability, coverage capability, and task efficiency [4]. Compared with single-UAV operation, multi-UAV collaboration exhibits distinct merits. It enhances system robustness through inter-agent fault tolerance, expands operational coverage via distributed deployment, and supports flexible scalability by adaptively adjusting the swarm scale and layout to satisfy task demands [5], [6]. Benefiting from these superior characteristics, multi-UAV systems are well suited for large-area reconnaissance, search and rescue, and disaster relief missions, where operational robustness and environmental adaptability are highly required [7], [8], [9], [10].

Task planning is regarded as the core link of multi-UAV cooperative coordination. By jointly optimizing flight trajectories, task allocation and communication strategies, cooperative planning enables multiple UAVs to operate as an integrated system for accomplishing complex mission objectives [11]. Before mission execution, the ground station formulates initial task and path arrangements for all UAVs, which is defined as task pre-planning. Based on global environmental and operational feedback uploaded by UAVs, pre-planning generates reasonable task sequences for each individual UAV under predefined mission constraints. Nevertheless, rigid pre-planning still exhibits obvious deficiencies in dynamic reconnaissance tasks, since it is highly vulnerable to task fluctuations, sensor faults and unexpected UAV damage [12]. Various uncertain factors in dynamic scenarios inevitably bring severe challenges to conventional pre-planning methods, among which two key limitations are

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particularly prominent. On one hand, most existing UAV cooperative frameworks adopt fixed communication topologies, lacking sufficient flexibility to adapt to time-varying task demands. Such limitation restricts the practical application scope of multi-UAV cooperation and hinders efficient collaborative execution of complex tasks [13]. On the other hand, multi-UAV cooperative missions are usually carried out in highly dynamic environments, accompanied by random task addition or removal and unexpected UAV node failure. These dynamic disturbances are prone to invalidate pre-scheduled plans, making traditional pre-planning difficult to maintain stable performance in practical execution [14]. Newly added or cancelled task targets require cross-UAV resource reallocation and task resequencing, while accidental UAV malfunction forces remaining healthy UAVs to undertake interrupted tasks and reorganize the overall plan, which imposes great difficulties on conventional static pre-planning strategies. To tackle the aforementioned challenges, this study proposes a Temporal State-aware Multi-UAV Cooperative Task Replanning (TSM-CTR) framework. The fundamental principle is to tightly couple temporal task sequencing with trajectory-aware planning, while adopting a distributed communication architecture to support scalable local decision-making. The proposed framework consists of three core components: (i) a dynamic clustering model with cluster head adaptation to sustain robust intra-cluster and inter-cluster communication in the face of changing network conditions; (ii) a hierarchical incremental replanning strategy that enables localized task adjustments at multiple levels, thereby reducing reliance on computationally expensive global re-optimization; and (iii) a locality-conscious orchestration mechanism that respects temporal constraints, UAV states, and resource limitations to ensure feasible and efficient task execution.

By emphasizing temporal state awareness and local decision optimization, TSM-CTR achieves superior operational robustness and dynamic responsiveness to adapt to complex and uncertain mission environments. The main contributions of this work are summarized as follows:

- A temporal state-aware cooperative task replanning mechanism that jointly considers UAV trajectories, states, and resource factors to dynamically reconstruct task sequences.
- A hybrid centralized–distributed communication architecture enabling direct intra-cluster communication and inter-cluster information exchange among cluster heads.
- A dynamic clustering and cluster head adaptation mechanism within the hybrid architecture, together with a hierarchical incremental replanning strategy addressing node and task changes in dynamic environments.

II. RELATED WORKS

Coordinated operation of multiple UAVs through distributed sensing, computing, and control significantly enhances robustness, coverage, and task throughput, making it a major research focus [15], [16], [17]. In terms of task allocation and resource constraints, existing works have demonstrated the effectiveness of distributed task allocation and time-window-constrained autonomous task allocation strategies, such as time-window constrained task allocation in multi-UAV systems [15], and biomimicry-based adaptive task allocation methods [16]. Moreover, a distributed task redistribution framework has been proposed for dynamic environments [17]. These works collectively reveal the crucial role of task allocation in improving system throughput and robustness [18]. Market methods have also been widely studied for dynamic task reassignment. NI-MTA implements a contract-net-inspired mechanism with task announcement, bid collection, and coalition selection to support dynamic allocation under communication constraints [19].

Dynamic task replanning and real-time collaborative planning are core capabilities for achieving efficient multi-UAV collaboration. However, task replanning in multi-UAV systems faces two key challenges. First, many approaches trigger task reordering only after a replanning request is received at the base station, then reorganize the task sequence based on the UAVs' current states, often neglecting the mapping between the execution sequence and flight trajectories, limiting applicability in practice [20]. In practice, flight trajectories are tightly coupled with the execution sequence, and ignoring this coupling can lead to scheduling inaccuracies and even conflicts or delays. Second, traditional replanning algorithms typically update global information after UAVs submit a replanning request and then perform a global adjustment using all UAVs and tasks to cope with sudden changes [21], but these methods incur high computational complexity and large time overhead, rendering real-time replanning in large-scale UAV clusters challenging [22].

In the context of large-scale, multi-cluster structured hybrid distributed-centralized architectures and methods (e.g., cluster head adaptation and dynamic clustering), these approaches offer effective means to enhance cooperative efficiency and scalability [23]. Researchers have proposed hybrid architectures that combine distributed control with centralized coordination, enabling efficient management and rapid task- and resource-response through dynamic clustering, cluster head adaptation, and multi-level replanning [24], [25]. These works emphasize maintaining robust and scalable information flow, task allocation, and resource scheduling in dynamic topologies and cross-cluster collaboration, with clear relevance to practical multi-UAV applications [26], [27]. Nevertheless, current research still faces challenges such as the coupling between trajectories and time series, the trade-off between local replanning and global coordination, and the real-time requirements of highly dynamic, resource-constrained environments. Further theoretical and algorithmic innovations are needed to enhance system adaptability and robustness [28].

III. SYSTEM MODEL AND PROBLEM FORMULATION

We consider a team of M UAVs $\mathcal{U} = \{U_1, \ldots, U_M\}$ and a set of N reconnaissance tasks $\mathcal{T} = \{T_1, \ldots, T_N\}$. Each UAV U_j is described by

$$U_j = \langle loc_j,\ E_j^{\max},\ F_j,\ V_j \rangle, \tag{1}$$

where $loc_j(t) \in \mathbb{R}^3$ is the position of U_j at time t (abbreviated as loc_j), $E_j^{\max} > 0$ is its maximum energy. $F_j \in \{0, 1\}$ is a binary state indicator: $F_j = 1$ means UAV U_j operates normally, while $F_j = 0$ means UAV U_j is faulty and cannot work properly. $V_j > 0$ is its maximum flight speed. Each task T_i is described by

$$T_i = \langle l_i,\ lrt_i,\ V_i,\ E_i \rangle, \tag{2}$$

where $l_i \in \mathbb{R}^3$ denotes the 3D location of the task target, lrt_i represents the latest allowable completion time (deadline) for the task, $V_i > 0$ is the reconnaissance value obtained by completing the task, and $E_i > 0$ indicates the energy required to execute the task.

We further define key notations for task sequence and performance metrics: P_j denotes the ordered task sequence assigned to UAV U_j, with $|P_j|$ representing the length of P_j (i.e., the number of tasks assigned to U_j), $p_{j,l}$ refers to the l-th task in the sequence P_j, $x_{i,j} \in \{0, 1\}$ is a binary task allocation variable where $x_{i,j} = 1$ if task T_i is assigned to UAV U_j and $x_{i,j} = 0$ otherwise, D stands for the total flight distance of all UAVs, and U denotes the overall task utility. To capture the dynamic spatial-temporal evolution of UAV positions under task replanning and node state changes, we introduce the following temporal state-aware function to model the time-varying location of each UAV.

Definition 1 (Temporal State-Aware Function): The real-time position of UAV U_j at time t is characterized by a temporal state-aware mapping function $f_j : t \rightarrow loc_j$. Accordingly, the 3D coordinate of U_j is formulated as $loc_j = [x_j', y_j', z_j']^T \in \mathbb{R}^3$.

$$D = \sum_{j=1}^{M} \sum_{l=1}^{|P_j|-1} \mathrm{dis}(p_{j,l}, p_{j,l+1}) \tag{3}$$

$$U = \sum_{i=1}^{N} V_i \cdot \left(\sum_{j=1}^{M} x_{i,j} \cdot F_j \right) - \sum_{i=1}^{N} \sum_{j=1}^{M} x_{i,j} \cdot E_i \tag{4}$$

Eq. (3) calculates the total flight distance D of all UAVs. Here, j indexes UAVs ($1 \leq j \leq M$), and l is the index defining the order of tasks in the sequence assigned to UAV U_j. P_j denotes the task sequence of UAV U_j, with $|P_j|$ being the total number of ordered task points in this sequence. $p_{j,l}$ and $p_{j,l+1}$ represent the l-th and $(l+1)$-th ordered task points in the sequence, respectively. The function $\mathrm{dis}(\cdot, \cdot)$ computes the Euclidean distance between their 3D coordinates. Specifically, the real-time UAV position loc_j is treated as a virtual task point and appended to P_j to compute the flight distance between consecutive ordered tasks.

Eq. (4) defines the system-level task utility U, which is formulated as the net profit of total task revenue minus total execution energy consumption. The first term represents the cumulative reconnaissance value obtained by all normally operating UAVs, where V_i denotes the inherent reward value of task T_i. The second term indicates the total energy consumption for task execution, where only tasks with $x_{i,j} = 1$ consume the corresponding execution energy E_i of T_i.

Constraints for feasible and efficient task allocation and UAV operation are as follows:

$$\sigma \cdot \sum_{l=1}^{|P_j|-1} \mathrm{dis}(p_{j,l}, p_{j,l+1}) + \sum_{l=1}^{|P_j|} E_{j,l} \leq E_j^{\max}, \quad j \in \{1, \ldots, M\} \tag{5}$$

Eq. (5) describes the onboard energy limitation for each UAV U_j. σ: UAV energy consumption rate per unit flight distance. $\sigma \cdot \sum_{l=1}^{|P_j|-1} \mathrm{dis}(p_{j,l}, p_{j,l+1})$: total flight energy consumption, which is consistent with the distance calculation in Eq. (3). $\sum_{l=1}^{|P_j|} E_{j,l}$: total energy consumed to finish all tasks in sequence P_j, where $E_{j,l} = E_i$ if task $p_{j,l}$ corresponds to T_i. The sum of flight energy and task execution energy is constrained within the maximum onboard energy capacity $E_j^{\max}$ of U_j.

In this study, we focused on the energy required for flight propulsion and mission execution, as these factors account for the majority of the energy consumption of the UAV in reconnaissance tasks. The energy consumption in communication and computing aspects is relatively small, so we made reasonable simplifications in this part.

$$\sum_{i=1}^{N} \sum_{j=1}^{M} x_{i,j} = N \tag{6}$$

Eq. (6) guarantees full task coverage. The summation of all elements in the allocation matrix $X = \{x_{i,j}\}_{N \times M}$ equals the total task number N, which ensures zero unassigned tasks.

$$\sum_{j=1}^{M} x_{i,j} = 1, \qquad i = 1, \ldots, N \tag{7}$$

Eq. (7) enforces exclusive task allocation. Each row summation of matrix X equals 1, ensuring that each task T_i is assigned to exactly one UAV.

Collectively, these constraints define a feasible multi-UAV task allocation problem under energy, coverage, and exclusivity requirements. Building on this system model, we next introduce the proposed TSM-CTR framework, which addresses dynamic task replanning under unexpected events.

IV. FRAMEWORK DESCRIPTION: TSM-CTR

The proposed framework integrates temporal sequencing with trajectory-aware planning under a distributed–centralized hybrid architecture, which consists of three tightly coupled core modules: a hybrid distributed–centralized dynamic clustering architecture, a dynamic distance-based clustering mechanism, and an incremental replanning strategy.

A. HYBRID DISTRIBUTED-CENTRALIZED DYNAMIC CLUSTERING ARCHITECTURE

This section proposes a three-layer hybrid control framework that integrates centralized coordination with distributed execution via dynamic clustering. To address the key challenges in multi-UAV cooperative reconnaissance—including poor scalability of purely centralized control, lack of global coordination in fully distributed systems, and high communication overhead under dynamic task changes and node failures—this hybrid architecture is designed to leverage the advantages of both paradigms. As illustrated in **Figure 1**, the ground control layer maintains reconnaissance data and conducts global cluster head elections. The task command layer consists of a dynamic set of cluster heads $G = \{G_1, \ldots, G_m\}$, each coordinating its cluster members and interfacing with other heads for inter-cluster coordination. The task execution layer comprises cluster members $H = \{H_b^a\}$, where H_b^a denotes the b-th member in the cluster headed by G_a and $G_a \in G$. The framework is succinctly represented by the triplet $\langle B, G, H \rangle$, with $G \subseteq U$ and $H_b^a \in H$.

A cluster head coordinates cluster planning and may execute tasks when appropriate. This hybrid architecture combines centralized oversight with distributed execution to balance scalability and robustness (see Section IV-C for explicit design assumptions). The ground control, task command and task execution layers share responsibilities to support resilient operation under dynamic conditions.

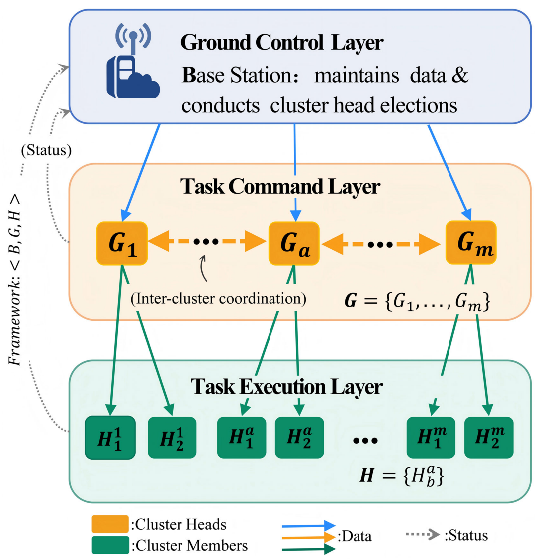

FIGURE 1. Schematic diagram of distributed control architecture.

B. DYNAMIC DISTANCE-BASED CLUSTERING

The dynamic distance-based clustering method for multi-UAV systems includes three main phases: cluster formation, cluster head selection, and cluster maintenance, as shown in **Figure 2**. In dynamic distance-based clustering, the multi-UAV network is partitioned into clusters according to velocity and distance similarity between UAVs. Importantly, cluster sizes must be balanced to prevent performance degradation. Thus, after initial clustering based on velocity and distance similarity, cluster sizes are validated by imposing a maximum allowable node limit $n_{\max}$ per cluster. The detailed procedure for dynamic distance-based clustering is outlined in **Algorithm 1**

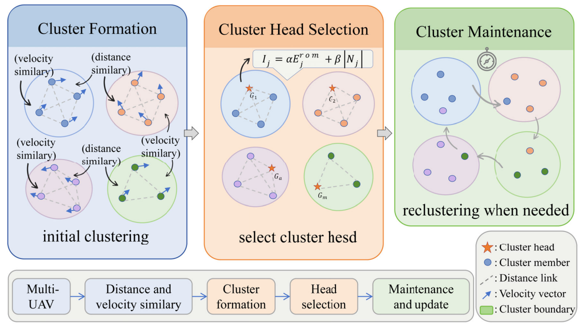

FIGURE 2. Dynamic distance clustering diagram.

For any UAV U_j and U_g in the network, their 3D positions at time t_1 and t_2 are denoted as $\boldsymbol{loc}_j^{t_1} = [x_j^{t_1}, y_j^{t_1}, z_j^{t_1}]^T$, $\boldsymbol{loc}_g^{t_1} = [x_g^{t_1}, y_g^{t_1}, z_g^{t_1}]^T$, $\boldsymbol{loc}_j^{t_2} = [x_j^{t_2}, y_j^{t_2}, z_j^{t_2}]^T$, and $\boldsymbol{loc}_g^{t_2} = [x_g^{t_2}, y_g^{t_2}, z_g^{t_2}]^T$, respectively. Let $\Delta t = t_2 - t_1$ denote the time period between two consecutive UAV position sampling instants, a fixed sampling interval used to compute average velocity and displacement for dynamic clustering. Assuming uniform motion of UAVs during reconnaissance, their average velocities are calculated as $\boldsymbol{v}_j = \frac{\boldsymbol{loc}_j^{t_2} - \boldsymbol{loc}_j^{t_1}}{\Delta t}$ and $\boldsymbol{v}_g = \frac{\boldsymbol{loc}_g^{t_2} - \boldsymbol{loc}_g^{t_1}}{\Delta t}$.

The velocity difference between UAV U_j and its neighbor U_g is defined as $\Delta v_{jg} = \|v_j - v_g\|$. The average velocity difference of U_j relative to all its neighbors is $\Delta v_j = \frac{1}{|N_j|}\sum_{g \in N_j} \Delta v_{jg}$, where N_j denotes the number of candidate neighbors of UAV U_j.

The distance difference over the time interval Δt is defined as the difference in displacement distances of U_j and U_g, given by $d_j^g = \left\| \boldsymbol{loc}_j^{t_2} - \boldsymbol{loc}_j^{t_1} \right\| - \left\| \boldsymbol{loc}_g^{t_2} - \boldsymbol{loc}_g^{t_1} \right\|$. Correspondingly, the average distance difference of UAV U_j relative to all its candidate neighbors is $\Delta d_j = \frac{1}{N_j} \sum_{g=1}^{N_j} d_j^g$.

Residual travel capability (residual energy) and node degree (neighbor count) are adopted as core evaluation metrics for cluster head selection. The residual energy of UAV U_j is calculated as:

$$E_j^{\mathrm{rem}} = E_j^{\max} - \sum_{l=1}^{|P_j|-1} \sigma \cdot \mathrm{dis}(p_{j,l}, p_{j,l+1}) - \sum_{l=1}^{|P_j|} E_{j,l} \qquad (8)$$

where $E_j^{\max}$ denotes the maximum energy capacity of UAV U_j, σ is the energy consumption coefficient per unit

Algorithm 1 Dynamic Distance-Based Clustering Algorithm

Require: Set of UAVs $\mathcal{U}$, thresholds q, p, max cluster size $n_{\max}$, period T, weights α, β
Ensure: Cluster head set G, cluster partition $\{H^a\}$
$G \leftarrow \varnothing$, $\{H^a\} \leftarrow \varnothing$, $t \leftarrow$ now(), $next\_update \leftarrow t + T$
while true **do**
  $t \leftarrow$ now()
  if $t < next\_update$ **then**
    sleep_until($next\_update$)
  end if
  for all $U_j \in \mathcal{U}$ **do**
    $N_j \leftarrow \varnothing$; obtain $\boldsymbol{v}_j$, $\boldsymbol{loc}_j^{t_1,t_2}$ from samples
    for all $U_g \in \mathcal{U} \setminus \{U_j\}$ **do**
      $\Delta v_{jg} \leftarrow \|\boldsymbol{v}_j - \boldsymbol{v}_g\|$, $d_{jg} \leftarrow \|\boldsymbol{loc}_j^{t_2} - \boldsymbol{loc}_j^{t_1}\| - \|\boldsymbol{loc}_g^{t_2} - \boldsymbol{loc}_g^{t_1}\|$
      if $\Delta v_{jg} \leq q$**and**$|d_{jg}| \leq p$ **then**
        $N_j \leftarrow N_j \cup \{U_g\}$
      end if
    end for
    $I_j \leftarrow \alpha E_j^{\text{rem}} + \beta |N_j|$
  end for
  build graph $\mathcal{G}_N(\mathcal{U}, E)$ with $(j, g) \in E \iff g \in N_j$; extract connected components $\{H^a\}$
  for all $H^a \in \{H^a\}$ **do**
    while $|H^a| > n_{\max}$ **do**
      compute P_j for $U_j \in H^a$; $U_{\text{prune}} \leftarrow \arg\max_j P_j$; $H^a \leftarrow H^a \setminus \{U_{\text{prune}}\}$
    end while
    $G_a \leftarrow \arg\max_{U_j \in H^a} I_j$; $G \leftarrow G \cup \{G_a\}$; $H^a \leftarrow H^a \setminus \{G_a\}$
  end for
  $next\_update \leftarrow t + T$
end while
return G, $\{H^a\}$

distance, $\text{dis}(p_{j,l}, p_{j,l+1})$ represents the Euclidean distance between consecutive waypoints $p_{j,l}$ and $p_{j,l+1}$, and $E_{j,l}$ is the communication energy consumption at waypoint $p_{j,l}$.

The joint evaluation metric for cluster head candidacy is $I(U_j) = \alpha \cdot E_j^{\text{rem}} + \beta \cdot |N_j|$. Here, α and β are tunable weight coefficients satisfying $\alpha + \beta = 1$, which adjust the relative importance of residual energy and node degree, respectively. Their values are determined based on practical mission requirements: a larger α is used to prioritize energy sustainability for long-endurance operations, while a larger β is assigned to enhance network connectivity in dense swarm scenarios. If the size of a cluster H^a exceeds the maximum allowable limit $n_{\max}$, nodes with the largest value of $(\Delta v_{jg} - \Delta d_{jg})$ are pruned sequentially until $|H^a| \leq n_{\max}$. The cluster head G_a is selected as the node with the maximum evaluation index $I(U_j)$ in the cluster, and the remaining nodes form the cluster member set H^a.

Periodic maintenance is governed by a clock period T. In the absence of task reassignments, dynamic distance-based clustering runs periodically with period T.When task reassignment occurs, the current time becomes the new maintenance reference, which is updated again upon subsequent reassignments.

Finally, in cooperative multi-UAV tasks, replanning strategies are deployed to address task additions/removals, UAV energy depletion, or malfunctions. The system identifies the triggering scenario and selects an appropriate replanning strategy, ensuring feasible and efficient task execution under dynamic conditions.

C. DESIGN CHOICES AND ASSUMPTIONS

To ensure practicality and reproducibility, we introduce several basic design choices and simplifying assumptions in this work.

(1) A hierarchical incremental replanning policy coordinated by cluster heads prioritizes local, low-overhead task adjustments before escalating to global replanning.

(2) Temporal state aware trajectory mapping generates time-consistent task sequences and physically feasible flight paths instead of idealized straight-line trajectories.

(3) Regular communication among UAVs is assumed to maintain normal information exchange.

(4) Uniform speed and simplified energy models focus the evaluation on replanning performance.

D. INCREMENTAL REPLANNING STRATEGY

The incremental replanning strategy addresses dynamic changes in UAV states or task conditions by applying a hierarchical sequence of replanning steps, yielding a new task reallocation scheme with minimal disruption.

1) MULTILEVEL STRUCTURE OF THE INCREMENTAL REPLANNING STRATEGY

The proposed incremental replanning strategy adopts a four-level hierarchical structure, designed to progressively expand the replanning scope while minimizing task disruption and computational overhead. This multilevel mechanism follows a strict local-first principle, attempting to resolve disturbances at the lowest feasible level before escalating to higher levels.

Level 1 (Self-Evaluation Replanning): The cluster head independently assesses its remaining energy and task capacity to absorb disturbances without involving other UAVs.

Level 2 (Intra-Cluster Assistance): If the cluster head cannot resolve the disturbance alone, assistance is requested from all members within the same cluster, limiting replanning to maintain locality.

Level 3 (Inter-Cluster Assistance): If intra-cluster resources are insufficient, the cluster head coordinates with neighboring cluster heads for cross-cluster collaboration, expanding the scope while avoiding global involvement.

Level 4 (Global Replanning): As the final fallback, the ground station initiates global replanning, involving all UAVs and tasks to resolve large-scale disturbances.

This layered framework balances flexible adaptation and operational efficiency. It reduces participating drones and communication costs amid minor fluctuations, while maintaining strong stability in extreme emergencies. Stepwise incremental replanning can well respond to real-time changes in UAV conditions and task demands, forming adjusted task allocation plans with little operational disruption.

2) INCREMENTAL REPLANNING STRATEGY UNDER DAMAGED UAV

Consider a damaged UAV within a cooperative multi-UAV task. The incremental replanning process upon UAV U_i damage at time t is illustrated in **Figure 3**.

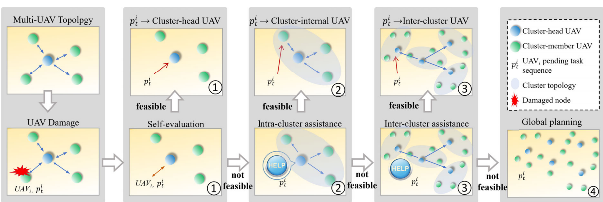

FIGURE 3. **Schematic diagram of incremental replanning strategy under *UAV$_i$* damage.**

Self-Evaluation Replanning (Module 1). When a cluster head detects that U_i is damaged, it assesses whether the remaining energy allows execution of the pending task sequence $p_i^{\prime}$ originally assigned to U_i. If feasible, $p_i^{\prime}$ is appended to the cluster head's own task sequence to compensate for the loss.

Intra-Cluster Assistance (Module 2). If the cluster head cannot execute $p_i^{\prime}$, it broadcasts a help request within the cluster. Each member evaluates whether it can execute $p_i^{\prime}$. If so, it absorbs $p_i^{\prime}$ into its pending sequence.

Inter-Cluster Assistance (Module 3). If intra-cluster support is insufficient, the cluster head requests assistance from neighboring cluster heads. Each receiving cluster performs its own self-evaluation and intra-cluster assistance to determine feasibility and returns results.

Global Replanning (Module 4). If all local steps fail, the ground base station intervenes to perform global replanning, or additional UAVs may be deployed to resolve the reassignment problem.

3) INCREMENTAL REPLANNING STRATEGY FOR NEWLY ADDED TASKS

When a new task appears at time t (denoted T_n), the following steps are taken (illustrated in **Figure 4**).

Self-Evaluation (Module 1). The cluster head evaluates whether it can accommodate T_n without compromising existing tasks. If feasible, T_n is inserted into its task sequence.

Intra-Cluster Assistance (Module 2). If not feasible for the head, the cluster head solicits assistance from other cluster members; a member that can execute T_n absorbs it into its pending sequence.

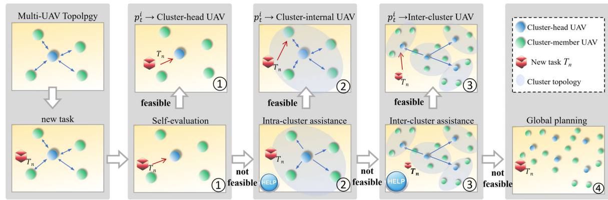

FIGURE 4. **Schematic diagram of incremental replanning strategy under newly added tasks.**

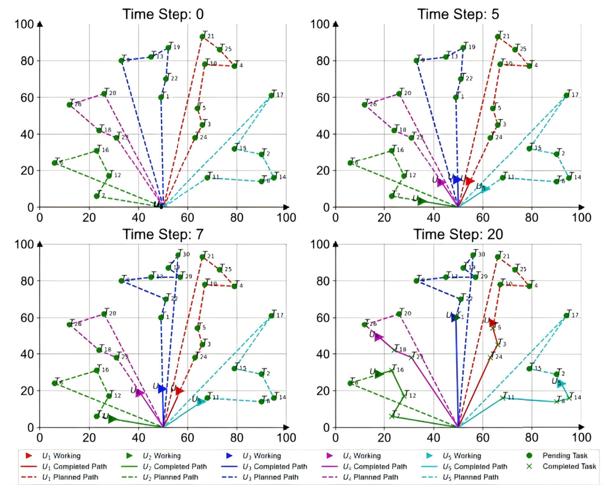

FIGURE 5. **Single machine replanning under newly added tasks.**

Inter-Cluster Assistance (Module 3). If intra-cluster help is insufficient, the head requests assistance from other clusters. Receiving cluster heads perform self-evaluation and intra-cluster assistance within their own cluster and report back.

Global Replanning (Module 4). If unresolved locally, global replanning is performed by the ground base station or by deploying additional UAVs to address the reassignment.

From the above, the incremental replanning strategies in the two scenarios share the same fundamental logic. Differences arise in triggering conditions and resource evaluation focus: newly added tasks emphasize matching additional resource requirements to available UAV resources, while UAV damage emphasizes redistribution of the remaining workload.

In UAV task replanning, the incremental strategy progresses through localized adjustments whenever possible. Only if local solutions are infeasible does the scope expand, potentially reaching global replanning. This approach offers local autonomy, reduced communication, and efficient resource utilization while preserving overall task robustness.

V. INCREMENTAL REPLANNING ALGORITHM

The incremental replanning framework adopts a hierarchical modular sequence to address dynamic variations in UAV states or task conditions. Each module first attempts local resolution before escalating to inter-module coordination,

Algorithm 2 New Task Set Generation Algorithm

Require: Cluster head UAV UAV_h, pending task sequence p_i', newly added task set T_n, number of clusters K, maximum iterations M

Ensure: New task set of UAV_h, T_h'

Initialize $C = \{C_1, \ldots, C_K\}$ by sampling K elements from $(p_i' \cup T_n)$
for $iter = 1$ to M **do**
  for each $s \in (p_i' \cup T_n)$ **do**
    $d_{\min} \leftarrow \infty$, $k_{\text{best}} \leftarrow -1$
    for $i = 1$ to K **do**
      $d \leftarrow \sqrt{\sum_{j=1}^{L}(s_j - C_{i,j})^2}$
      if $d < d_{\min}$ **then**
        $d_{\min} \leftarrow d$
        $k_{\text{best}} \leftarrow i$
      end if
    end for
    Assign s to $\mathcal{K}_{k_{\text{best}}}$
  end for
  $C_{\text{old}} \leftarrow C$
  for $i = 1$ to K **do**
    if $\mathcal{K}_i \neq \emptyset$ **then**
      $C_i \leftarrow \frac{1}{|\mathcal{K}_i|} \sum_{s \in \mathcal{K}_i} s$
    end if
  end for
  if $C = C_{\text{old}}$ **then**
    break
  end if
end for
return T_h'

thereby minimizing task disruption and preserving system robustness.

A. SELF-EVALUATION REPLANNING ALGORITHM

The self-evaluation module serves as the first layer of the hierarchical framework. Upon detecting a UAV damage or newly added tasks, the cluster head initiates self-evaluation to verify the feasibility of local task reassignment. If feasible, the proposed adjustments are integrated locally without further escalation. The detailed procedure is outlined in **Algorithm 2**.

This algorithm initializes K centroids from the union of the current task sequence and newly added tasks, where s denotes the task sample to be assigned, C_i is the i-th cluster centroid, L is the dimension of the task sample, and $\mathcal{K}_i$ represents the i-th task cluster. Meanwhile, $d_{\min}$ is the minimum Euclidean distance from the sample to each centroid, k_{best} is the optimal cluster index for sample assignment, and C_{old} is the centroid set of the previous iteration for convergence judgment. It then iteratively assigns each sample to the nearest centroid and updates centroids until convergence. The output is a new task set T_h' tailored for UAV_h. A single-cluster mode ($K = 1$) is adopted in this short-circuit step to enable rapid local

Algorithm 3 Task Sequence Optimization Algorithm

Require: UAV U_h, new task set T_h', maximum number of generations $G_{\max}$

Ensure: Feasible task sequence p_h' for U_h, or trigger intra-cluster reassignment

$\mathcal{P} \leftarrow$ initialize_population(T_h')
for $iter = 1$ to $G_{\max}$ **do**
  $\mathbf{f} \leftarrow$ evaluate_fitness($\mathcal{P}$)
  $\mathcal{P}_{\text{par}} \leftarrow$ select_parents($\mathcal{P}$, $\mathbf{f}$)
  $\mathcal{P}_{\text{off}} \leftarrow$ crossover($\mathcal{P}_{\text{par}}$)
  $\mathcal{P}_{\text{off}} \leftarrow$ mutate($\mathcal{P}_{\text{off}}$)
  $\mathcal{P} \leftarrow$ select_new_population($\mathcal{P}$, $\mathcal{P}_{\text{off}}$)
end for
$p_h^* \leftarrow$ get_best_sequence($\mathcal{P}$)
$E_{\text{req}} \leftarrow$ calculate_energy(p_h^*)
if $E_{\text{req}} \leq E_h^{\text{rem}}$ **then**
  return $p_h' \leftarrow p_h^*$
else
  Trigger intra-cluster assistance reassignment
end if

reassignment when feasible. The convergence condition is defined as:

$$C_i = C_{\text{old},i}, \quad \forall i \tag{9}$$

If local allocation of the new task set T_h' is infeasible, the task sequence optimization algorithm refines task allocation within the cluster. It searches for an energy-efficient and feasible task sequence p_h' for UAV_h. The procedure is detailed in **Algorithm 3**.

This algorithm employs an evolutionary optimization scheme to search for candidate task sequences from T_h', guided by the fitness function defined in Eq. 10. In this context, $\mathcal{P}$ denotes the candidate sequence population, $\mathbf{f}$ is the fitness vector of the population, p_h^* is the optimal task sequence obtained by evolutionary optimization, E_{req} is the energy required to execute p_h^*, and E_h^{rem} is the remaining energy of UAV U_h. The fitness function is formulated as:

$$fitness = \frac{1}{\sum_{a=1}^{N_{T_h'}} \sum_{b=1}^{N_{T_h'}} \text{dis}(T_a, T_b)\mu_{ab}} \tag{10}$$

where $N_{T_h'}$ denotes the number of task points in T_h', $\text{dis}(\cdot, \cdot)$ calculates the Euclidean distance between two task points, and $\mu_{ab} = 1$ if T_a and T_b are consecutive in the path (otherwise $\mu_{ab} = 0$). The search terminates at the maximum generation or upon convergence, yielding the optimal path p_h^* and its required energy E_{req}. If $E_{\text{req}} \leq E_h^{\text{rem}}$, the optimal sequence is selected as p_h'; otherwise, intra-cluster assistance reassignment is triggered.

B. INTRA-CLUSTER ASSISTANCE REPLANNING ALGORITHM

When the cluster head cannot resolve the dynamic variation independently, intra-cluster assistance is activated to solicit

Algorithm 4 Shortest-Distance-First Task Allocation Algorithm

Require: Subtask centroids $\mathcal{C}$, cluster UAV set $H_b^a \cup G^a$
Ensure: Task allocation matrix X_{alloc}

$X_{\text{alloc}} \leftarrow \emptyset$
$u^* \leftarrow \emptyset$
$d_{\min} \leftarrow \infty$
for each $c_p \in \mathcal{C}$ **do**
  for each $U_j \in H_b^a \cup G^a$ **do**
    $d \leftarrow \text{dis}(c_p, U_j)$
    if $d < d_{\min}$**and**$x_{p,j} = 0$ **then**
      $d_{\min} \leftarrow d$
      $u^* \leftarrow U_j$
    end if
  end for
  $x_{p,j} \leftarrow 1$
  $X_{\text{alloc}} \leftarrow X_{\text{alloc}} \cup \{x_{p,j}\}$
end for
return X_{alloc}

support from other cluster members. This intra-cluster process is equivalent to parallel self-evaluation replanning across multiple UAVs within the cluster; capable members absorb the new task into their pending sequences. If no member can undertake the task, the system proceeds to inter-cluster assistance.

Inter-cluster assistance enables cross-cluster collaboration: the cluster head selects potential executors from other clusters, which perform self-evaluation and intra-cluster assistance (if required) and report results back.

The shortest-distance-first allocation strategy for intra-cluster reassignment is described in **Algorithm 4**. Subtasks are allocated to UAVs within the cluster based on the nearest-centroid criterion. In Algorithm 4, c_p denotes the p-th subtask centroid, u^* is the nearest available UAV, $d_{\min}$ is the minimum distance, and X_{alloc} is the task allocation matrix. The resulting X_{alloc} establishes a mapping between subtasks and executors, enabling fast local reallocation for minor cluster-level changes.

C. INTER-CLUSTER ASSISTANCE REPLANNING ALGORITHM

If intra-cluster resources remain insufficient, the cluster head initiates inter-cluster assistance. Through inter-cluster communication, neighboring clusters are queried to perform self-evaluation and intra-cluster assistance to determine task feasibility. Results are returned to the source cluster head to guide subsequent actions. If no feasible solution is obtained, full replanning is invoked.

D. COMPLETE REPLANNING ALGORITHM

When local and inter-cluster strategies fail, a global full reallocation is performed. The ground station or central controller collects status information from all clusters, generates a new global task set, and redistributes tasks using the shortest-distance principle. Each cluster head then re-executes the self-evaluation and intra-cluster allocation steps to finalize the plan. If certain UAVs cannot fulfill their assigned tasks, additional UAVs may be deployed or tasks reassigned to maintain overall task feasibility and robustness.

E. COMPLEXITY AND CONVERGENCE ANALYSIS

For the proposed hierarchical incremental replanning framework, the overall computational complexity is dominated by the task sequence optimization module, which adopts an evolutionary algorithm. The outer clustering and multi-level replanning modules exhibit linear complexity with respect to the number of UAVs and tasks, $O(M + N)$, which is computationally lightweight.

The evolutionary algorithm inherently introduces additional computational overhead, with a complexity bound of $O(G_{\max} \cdot P \cdot L)$, where $G_{\max}$ denotes the maximum number of iterations, P is the population size, and L represents the length of the task sequence. However, our hierarchical design significantly reduces the problem scale by limiting replanning to local clusters rather than the entire swarm, effectively bounding the complexity within a manageable range.

Regarding convergence, the evolutionary algorithm converges to a feasible suboptimal solution within a finite number of iterations under reasonable parameter settings, which is sufficient for real-time replanning requirements in dynamic reconnaissance scenarios. We acknowledge that a full theoretical convergence proof is non-trivial for evolutionary methods, and rigorous theoretical analysis will be pursued as an important direction in our future work.

VI. EXPERIMENTAL RESULTS AND ANALYSIS

This section evaluates the proposed TSM-CTR framework through two experimental parts. Part I conducts qualitative replanning simulations in both small-scale ($N = 28, M = 5$) and large-scale ($N = 40, M = 8$) scenarios under newly added tasks and UAV damage, illustrating the hierarchical incremental replanning mechanism in **Figures 5–13**. Part II performs quantitative evaluations including parameter sensitivity analysis and comprehensive baseline comparisons. Evaluations adopt system mission utility, replanning time, and the number of involved UAVs as core metrics, with results summarized in **Figures 14–16**.

A. EXPERIMENTAL SETUP

All experiments are implemented in Python 3.10 on a workstation equipped with an AMD Ryzen 7 5800H processor and 16 GB RAM. In the pre-planning phase, all UAVs are initialized at (50, 0), and the initial task allocation is visualized at Time Step = 0. For each experimental configuration, 500 independent trials are conducted to ensure statistical validity.

Simulation parameters for both scenario scales are presented in Table 1. The adapted baseline NI-MTA enforces practical constraints: a maximum of 5 bidders per task and a coalition size of ≤ 3.

TABLE 1. Simulation parameter settings.

Parameter	Value
Scales	$(N = 28, M = 5)$ and $(N = 40, M = 8)$
Initial UAV positions	All UAVs initialized at $(50, 0)$
Task value V_i	$V_i \in (20, 40)$
Task energy E_i	$E_i \in (10, 40)$
Latest reconnaissance time lrt_i	$lrt_i \in (0, 30)$

B. PART I: INCREMENTAL REPLANNING EXPERIMENT OF TSM-CTR

This section validates the effectiveness, adaptability, and robustness of the proposed TSM-CTR hierarchical incremental replanning mechanism. Two mission scenarios with progressive complexity are designed for sufficient evaluation, including a small-scale scenario ($N = 28, M = 5$) and an expanded large-scale scenario ($N = 40, M = 8$). Two typical dynamic disturbances in UAV cooperative reconnaissance missions are considered: Newly Added Tasks and UAV Damage.

Full-layer replanning logic of TSM-CTR is verified, covering single-UAV, intra-cluster, inter-cluster, and global optimization strategies. The small-scale scenario undertakes basic algorithm verification under low-load and simple disturbance conditions. In contrast, the expanded scenario adopts denser tasks and more UAV members. The experimental analysis focuses on replanning triggering rules, hierarchical escalation mechanisms, task sequence optimization, and flight trajectory adjustment effects.

1) INCREMENTAL REPLANNING UNDER NEWLY ADDED TASKS

This section evaluates the incremental replanning of TSM-CTR under dynamically added new tasks. It aims to verify the adaptive ability of the replanning mechanism for dynamic task changes. At Time Step = 7, cluster head U_3 detects two newly arrived tasks $\{T_{29}, T_{30}\}$ and triggers hierarchical replanning according to task deadline constraints.

If local task integration is feasible, U_3 directly incorporates $\{T_{29}, T_{30}\}$ into its existing schedule, yielding an updated task sequence $[T_1, T_{22}, T_9, T_{13}, T_{29}, T_{19}, T_{30}]$, as visualized in **Figure 5**. When individual scheduling fails to satisfy temporal constraints, intra-cluster collaboration among U_1, U_3, and U_4 is activated. The optimized task sequences of involved UAVs are $U_3 : [T_{20}, T_9, T_{13}]$ and $U_4 : [T_{24}, T_{23}, T_{18}, T_{26}]$, with corresponding trajectories demonstrated in **Figure 6**.

When intra-cluster resources are deficient, cross-cluster coordination is activated. The optimized task sequences for participating UAVs are $U_3 : [T_{24}, T_3, T_5, T_1, T_{22}]$, $U_4 : [T_{23}, T_{18}, T_{26}, T_{20}, T_9]$, and $U_1 : [T_{11}, T_{15}, T_2, T_8, T_{14}]$, and the adjusted flight paths are shown in **Figure 7**.

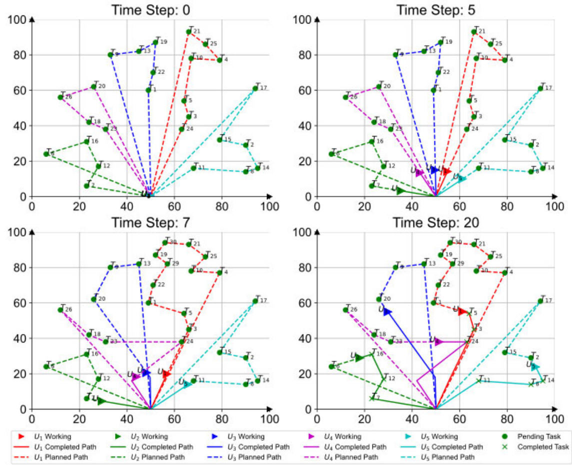

FIGURE 6. Intra-cluster replanning under newly added tasks.

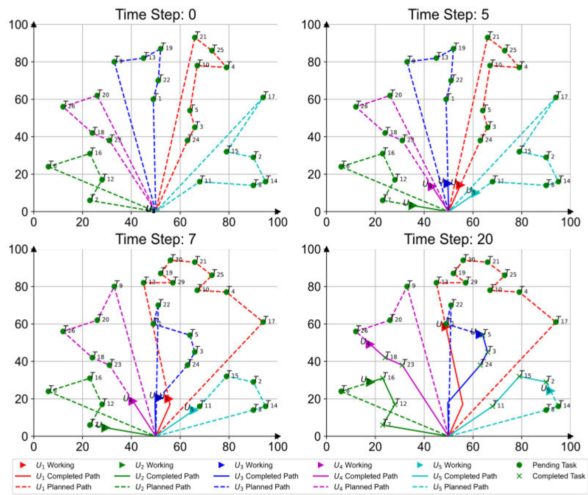

FIGURE 7. Inter-cluster replanning under newly added tasks.

If hierarchical adjustment cannot eliminate disturbances, global replanning is adopted to reallocate overall tasks. For instance, U_1's original sequence $[T_{24}, T_3, T_5, T_{10}, T_4, T_{25}, T_{21}]$ is updated to $[T_5, T_{10}, T_4, T_{25}, T_{21}, T_{30}, T_{19}, T_{29}]$ through network-wide path optimization, as shown in **Figure 8**.

The incremental framework achieves enhanced robustness and reduced task delays via localized-to-global hierarchical escalation. As illustrated in **Figure 5–8**, local replanning completes with minimal communication overhead when feasible, while hierarchical expansion preserves task feasibility without excessive resource consumption.

2) INCREMENTAL REPLANNING UNDER UAV DAMAGE

This subsection evaluates the incremental replanning performance of TSM-CTR under the scenario of single UAV damage, focusing on the system's resilience and recovery capability. In the experiment, U_3 is set to malfunction at Time Step = 7, which triggers the incremental replanning

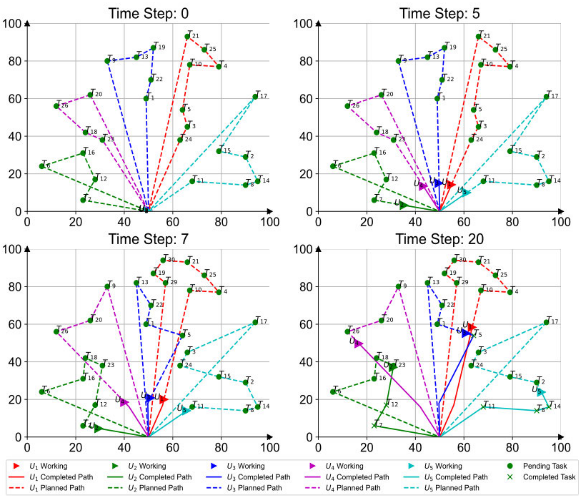

FIGURE 8. Global replanning under newly added tasks.

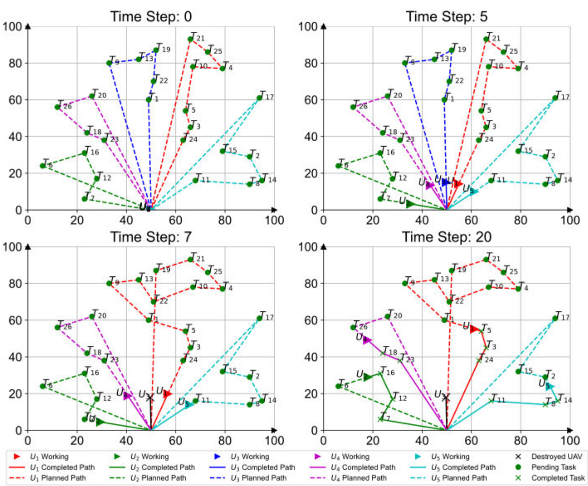

FIGURE 9. Single machine replanning under UAV damage.

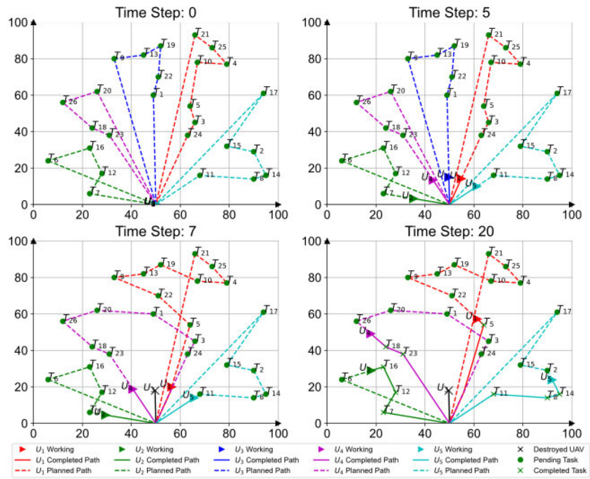

FIGURE 10. Intra-cluster replanning under UAV damage.

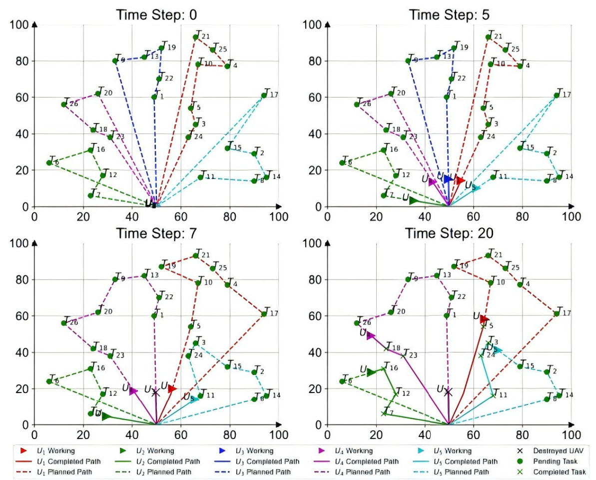

FIGURE 11. Inter-cluster replanning under UAV damage.

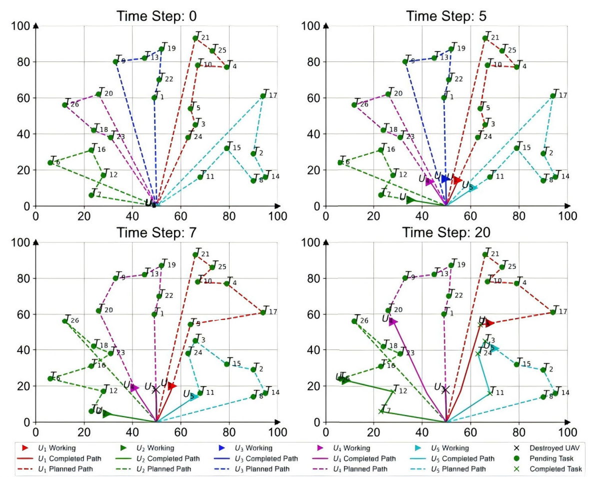

FIGURE 12. Global replanning under UAV damage.

mechanism to migrate unfinished tasks and restore swarm operational stability.

Cluster head U_1 first performs single-UAV replanning to accommodate residual tasks of U_3 and optimizes its local task sequence, as illustrated in **Figure 9**. If U_1 cannot meet task deadlines, intra-cluster collaboration with U_4 is triggered. The optimized task sequences of U_1 and U_4 are updated accordingly, and the corresponding trajectories are presented in **Figure 10**.

For intra-cluster resources insufficient, inter-cluster cooperation with U_5 is activated, yielding optimized task sequences (**Figure 11**).If all localized strategies fail, global replanning is invoked **Figure 12**. The updated schedules incorporate explicit temporal mapping of UAV positions, generating realistic trajectories rather than idealized straight-line paths.

The proposed incremental framework achieves graceful performance degradation under UAV damage. Localized adjustments minimize mission disruption, while global replanning is only invoked as a fallback strategy.

This hierarchical mechanism effectively enhances system resilience while restraining computational and communication overhead.

3) EXTENDED VERIFICATION IN LARGE-SCALE COMPLEX SCENARIO

This subsection further validates the incremental replanning performance of the TSM-CTR algorithm under complex high-load conditions. A large-scale complex scenario ($N = 40, M = 8$) is constructed with increased task density and UAV quantity, as well as compound dynamic disturbances, to comprehensively evaluate the algorithm's environmental adaptability and anti-disturbance robustness.

The large-scale scenario is configured with 38 initial reconnaissance tasks (T_1–T_{38}) and 7 UAVs. Three typical dynamic disturbances are sequentially introduced during mission execution: U_1 is damaged and exits the mission at Time Step = 7; two new tasks T_{39} and T_{40} are dynamically inserted at Time Step = 9; U_7 suffers secondary damage at Time Step = 15. The superposition of multi-stage UAV damage and delayed task increment forms compound disturbances, which significantly improves mission complexity.

TSM-CTR performs hierarchical incremental replanning following the unified logic of single-machine optimization, intra-cluster collaboration, inter-cluster coordination, and global optimization. After the damage of U_1, local cluster scheduling is preferentially activated to undertake residual tasks. When new incremental tasks arrive, the algorithm adaptively integrates tasks and triggers hierarchical escalation once local resources are saturated. Facing the subsequent failure of U_7, cross-cluster load balancing and global optimization are further implemented to maintain stable mission execution.

The typical replanning trajectories and optimization results of the TSM-CTR under the large-scale compound disturbance scenario are illustrated in **Figure 13**, which respectively demonstrate the intermediate adaptive adjustment effect after task increment and multi-UAV damage, as well as the final global optimized trajectory.

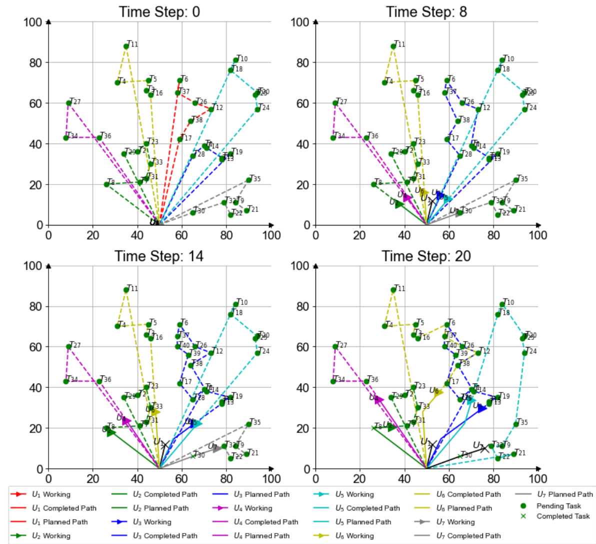

FIGURE 13. Incremental replanning under compound disturbances of newly added tasks and UAV damage.

C. PART II: COMPARATIVE EVALUATION OF TSM-CTR REPLANNING PERFORMANCE

Different from the qualitative case analysis of TSM-CTR in Part I, this section presents comparative experiments under two network scales. Three baseline methods are introduced for comparison, and two key metrics, including system mission utility and the number of UAVs involved in replanning, are used to evaluate the overall performance of the TSM-CTR.

1) EXPERIMENTAL BASELINES AND DYNAMIC SCENARIO SETTINGS

Three typical UAV scheduling methods covering global optimization, threshold-based hierarchical adjustment, and market negotiation are selected as baselines for comprehensive performance evaluation. The detailed descriptions of comparative methods are presented below:

- NI-MTA [19]: A contract-net-based market-driven method that realizes dynamic task allocation via task publishing, bidding selection, and coalition optimization under communication constraints. It is adapted to the UAV dynamic replanning framework as a market-strategy baseline.
- FCM-ASO [20]: A node variation awareness hierarchical scheduling method with fixed thresholds. It adopts local adjustment for node variation below 20%, intra-group adjustment for 20%–50% variation, and global adjustment for variation exceeding 50%.
- Global Replanning [29]: A classic dynamic scheduling strategy. Once disturbances occur, all UAVs participate in global task reallocation, serving as the baseline for full optimization and maximum resource overhead.

To validate the performance of TSM-CTR, comparative experiments are performed under two network scales, namely ($N = 28, M = 5$) and ($N = 40, M = 8$). This subsection compares the proposed TSM-CTR method with three baseline schemes in dynamic scenarios, where four distinct experimental scenarios are designed and eight groups of disturbance conditions considering newly added tasks and damaged UAVs are adopted for both network settings; the detailed configurations are listed in Table 2.

2) SENSITIVITY ANALYSIS OF CLUSTER MAINTENANCE PERIOD T

This section analyzes the sensitivity of the proposed TSM-CTR algorithm to the cluster maintenance period T, aiming to identify the optimal setting for balancing mission performance and system overhead.

TABLE 2. Scenario settings.

Scenario ID	N_{Task}	N_{UAV}	Dynamic Settings	
			New Tasks	Damaged UAVs
1	26	5	2	0
2	28	6	5	0
3	26	5	0	1
4	28	6	0	2
5	38	7	3	0
6	40	8	6	0
7	38	7	2	2
8	40	8	1	3

The experiment is conducted under Scenario 7, with T varying from 0 to 30 time steps. Two metrics are evaluated: average system mission utility and the number of UAVs involved in replanning.

As shown in **Figure 14**, the algorithm achieves the highest mission utility of 1770 at $T = 5$, which is 44.7% higher than the no-maintenance baseline ($T = 0$). As T increases, the number of UAVs involved in replanning gradually decreases from 6 to 4. When T is excessively small ($T < 5$), frequent maintenance introduces high communication and computational overhead, reducing the overall utility. When T exceeds 10, delayed cluster updates lead to a 14.0% drop in mission utility compared with the peak value.

Consequently, the optimal maintenance period is set to $T = 5$ time steps, which achieves a favorable trade-off between replanning optimization performance and system resource consumption.

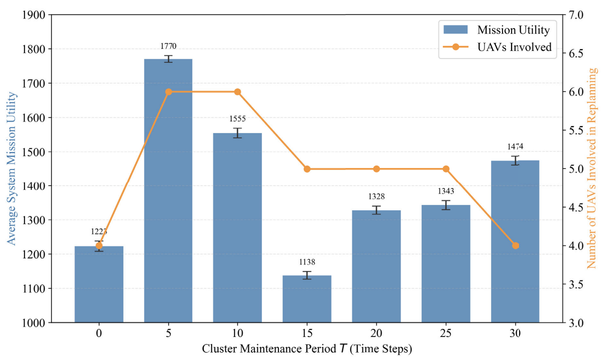

FIGURE 14. Sensitivity analysis of cluster maintenance period T.

3) UTILITY PERFORMANCE COMPARISON ACROSS ALL SCENARIOS

Comparative experiments covering eight predefined scenarios are carried out to assess the optimization performance of TSM-CTR under diverse disturbance intensities and mission scales. Four algorithms are adopted for benchmark comparison, namely NI-MTA, FCM-ASO, Global Replanning and the proposed TSM-CTR. Each scenario undergoes 500 independent simulations, and averaged mission utility is extracted for quantitative evaluation.

The utility comparison results are illustrated in **Figure 15**. TSM-CTR consistently achieves the highest mission utility across all scenarios, and its advantage becomes increasingly prominent as the task scale and disturbance complexity increase.

In small-scale scenarios (S1–S4), where the number of initial tasks is limited, all algorithms perform closely, with TSM-CTR yielding a moderate improvement of 3.6%–6.7% over the best baseline. In medium- and large-scale scenarios (S5–S8), involving more tasks and compound disturbances, the performance gap widens noticeably. Scenario 6, with the largest number of newly inserted tasks, yields the highest overall mission utility, and TSM-CTR maintains a stable advantage of approximately 7.0% over the best baseline. In scenarios with UAV damage (S7–S8), TSM-CTR exhibits the slowest performance degradation, retaining a clear utility advantage under severe operational failures.

Global Replanning incurs excessive overhead from full reallocation. NI-MTA is limited by negotiation efficiency, while FCM-ASO relies on fixed thresholds with limited adaptability. In contrast, TSM-CTR dynamically identifies the minimum replanning scope, effectively balancing mission performance and scheduling overhead, especially in large-scale complex environments.

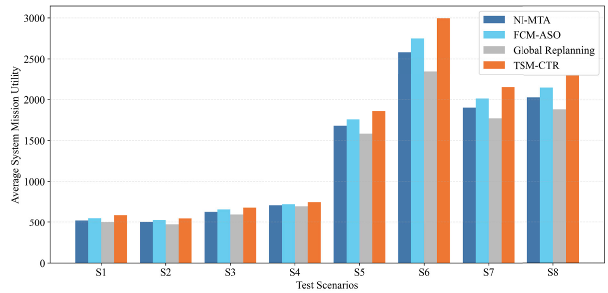

FIGURE 15. Mission utility comparison across eight test scenarios.

Performance differences across comparison methods stem from inherent scheduling mechanisms. Global Replanning triggers full-system task reallocation against disturbances, bringing severe resource redundancy and low mission utility. The market-driven NI-MTA suffers from limited negotiation efficiency and unsatisfactory task matching results under communication constraints. FCM-ASO adopts three-layer hierarchical adjustment to cut unnecessary scheduling consumption and obtains better performance than the above two methods, yet fixed threshold setting weakens its adaptability to dynamic swarm variations.

Different from baseline schemes, TSM-CTR utilizes temporal state-aware incremental replanning and selects the minimal effective adjustment scope matching actual disturbance intensity. This design effectively avoids excessive overhead in global optimization, remedies low efficiency of negotiation-based strategies and overcomes static limitation of threshold-driven rules, thus delivering superior and stable mission utility in complex dynamic scenarios.

4) REPLANNING TIME EFFICIENCY ANALYSIS

To further evaluate the real-time performance and computational overhead of the proposed TSM-CTR algorithm, replanning experiments are conducted under three communication ranges (150 m, 300 m, and 1000 m), with NI-MTA, FCM-ASO, and Global Replanning as baselines. The results, including average replanning time and the number of involved UAVs, are presented in **Figure 16**.

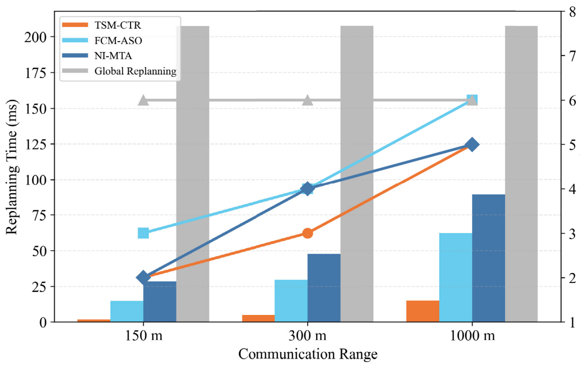

FIGURE 16. Replanning time and number of involved UAVs under different communication ranges.

As shown in the figure, TSM-CTR consistently outperforms all baseline methods in both metrics. In terms of replanning time, the proposed algorithm exhibits the lowest computational overhead under all communication conditions. Specifically, in the communication-constrained scenario (150 m), TSM-CTR requires only 1.71 ms per replan, which is 8.6 times faster than FCM-ASO, 16.6 times faster than NI-MTA, and 121.3 times faster than Global Replanning. As the communication range expands, the replanning time of TSM-CTR increases slightly but remains significantly lower than all baselines. Even under full communication coverage (1000 m), the replanning time of TSM-CTR is only 15.01 ms, which is still 4 times faster than FCM-ASO and 13.8 times faster than Global Replanning.

Correspondingly, the number of UAVs involved in replanning further confirms the efficiency advantage of TSM-CTR. At all communication ranges, Global Replanning involves all 6 UAVs, leading to persistent high overhead. As the communication range expands, both NI-MTA and FCM-ASO require more UAVs to participate, with FCM-ASO reaching 6 participants at 1000 m. In contrast, TSM-CTR maintains the smallest or equal participation scale: at 150 m, it matches NI-MTA at 2 UAVs; at 300 m, it remains equal to both NI-MTA and FCM-ASO at 4 UAVs; at 1000 m, it involves 5 UAVs, which is fewer than FCM-ASO and equal to NI-MTA. Throughout all scenarios, TSM-CTR never exceeds the participation scale of any baseline method.

These results demonstrate that TSM-CTR effectively reduces replanning scale and computational overhead through hierarchical cluster management and local incremental adjustment. Its advantage is particularly prominent in communication-constrained environments, strongly confirming the real-time capability and engineering practicability of the proposed algorithm in dynamic UAV cooperative missions with newly added tasks and UAV damage.

VII. CONCLUSION

This work presents a temporal state-aware multi-UAV cooperative task replanning framework for dynamic reconnaissance missions under unexpected events, including newly added tasks and UAV damage. A hybrid distributed–centralized communication architecture is developed, and a dynamic distance-based clustering mechanism is designed to support scalable and resilient swarm coordination. On this basis, a hierarchical incremental replanning strategy is proposed, consisting of self-evaluation, intra-cluster, inter-cluster, and global replanning, to adaptively respond to disturbances at different scales.

Simulation results across diverse scenarios demonstrate that the proposed method can dynamically adjust task sequences and UAV trajectories according to disturbance severity. By prioritizing local adjustments before invoking global optimization, the framework effectively reduces replanning overhead while maintaining stable task performance. Comparative experiments verify that TSM-CTR outperforms conventional global and fixed-threshold replanning methods in terms of system mission utility and scalability.

Future work will extend the current framework by incorporating practical communication constraints such as network latency and packet loss, introducing hovering and obstacle-avoidance motion models, and integrating communication/computation energy consumption metrics. In addition, rigorous convergence proof of the evolutionary-based optimization will be investigated to further enhance the practicability and theoretical rigor of the proposed approach.

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